Lecture 5 - P Control & System Visualization

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5.1 Example: P Control of 2nd Order Unstable Plant

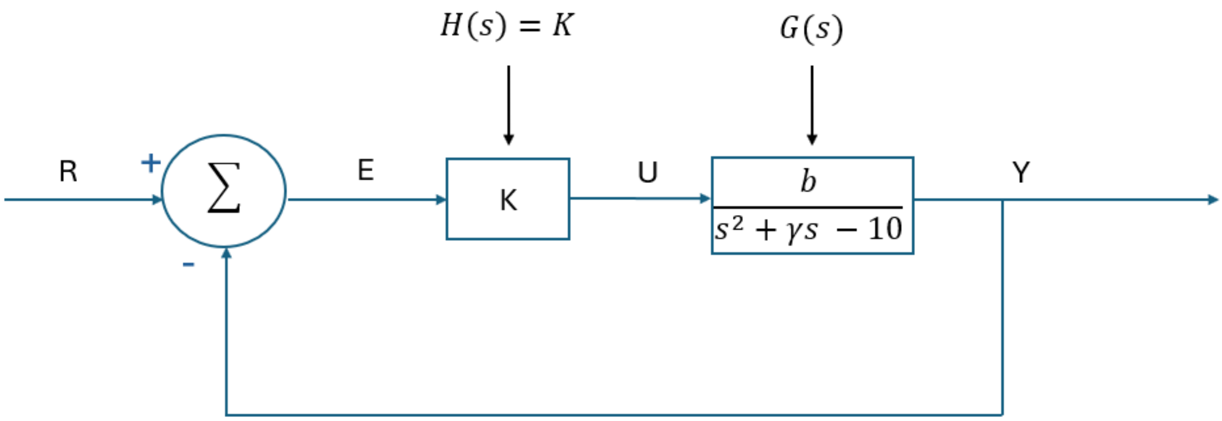

Figure 5.1: Block diagram for an unstable 2nd order plant and P controller.

Let $\gamma = 20$ and $b = 1$ in Figure 5.1 such that:

$$G_{ry}(s) = \frac{GH}{1+GH} = \frac{\frac{K}{s^2+20s-10}}{1+\frac{K}{s^2+20s-10}} = \frac{K}{s^2+20s-10+K}$$

Question: What values of K will stabilize G_{RY}?
$s^2 + a_1 s + a_2$: negative real roots iff $a_1, a_2 > 0$
[Ans: $-10 + K > 0 \implies K > 10$]

Question: What values of K have good tracking of a step input?
[Ans: $K >> 10$ such that $|\frac{K}{-10+K}| \approx 1$]

Question: Can we choose K to set the natural frequency $\omega_n = 5$?
[Ans: $\omega_n^2 = -10 + K = 25 \quad \rightarrow \quad K = 35$]

Question: Can we choose K to set the damping ratio?
[Ans: Yes, but not independently of ω_n]

5.2 Visualizing Responses Based on Poles

Reminder: Visualizing 1st order systems

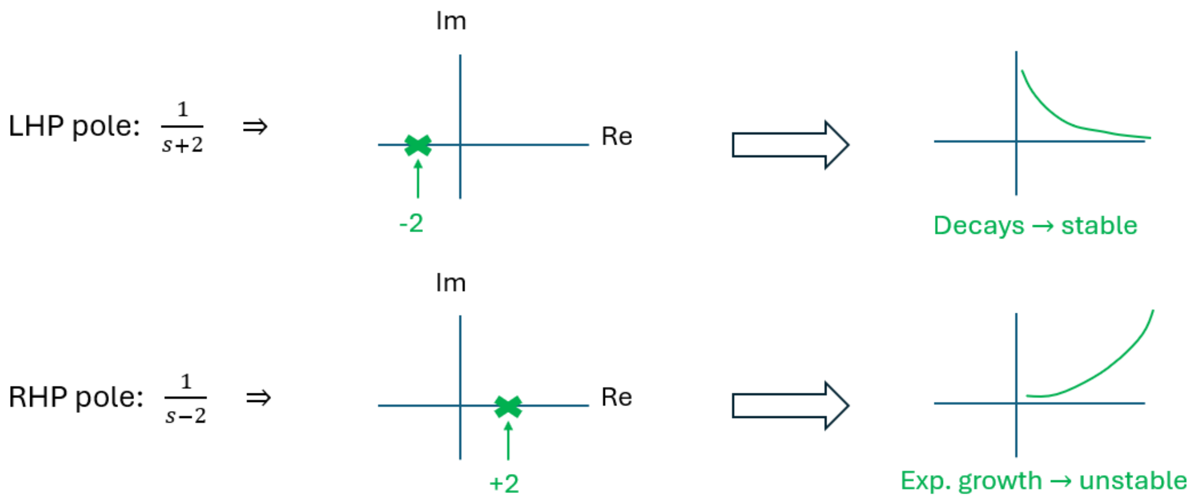

Figure 5.2: 1st order system response based on poles.

Visualizing 2nd Order Systems

Case 1: λ_1, λ_2 are real and negative (e.g. our Example 1 above)

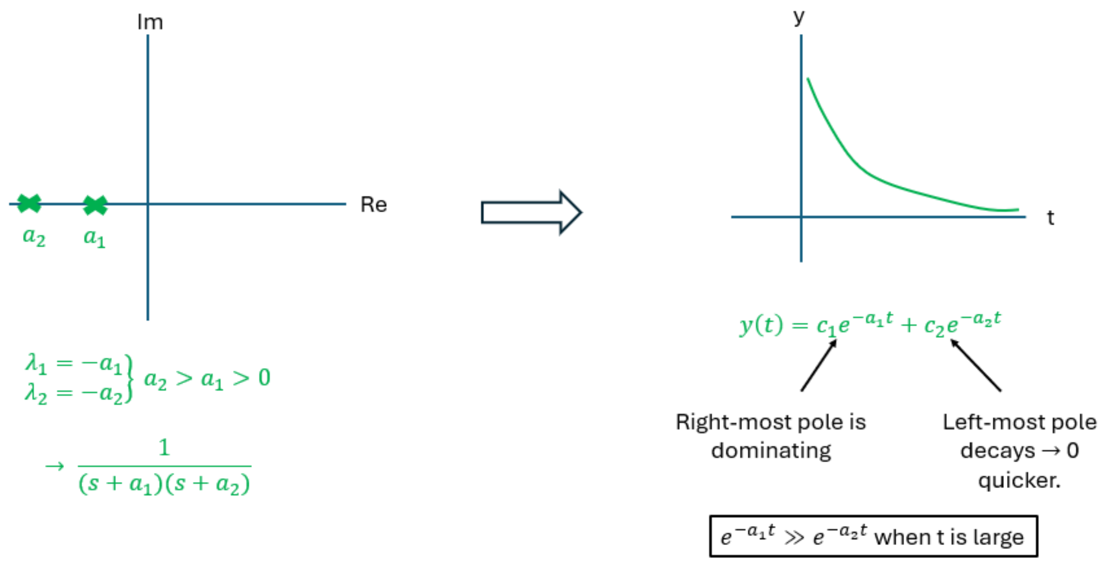

Figure 5.3: Second order system response for real and negative poles.

Case 2: λ_1, λ_2 are purely imaginary

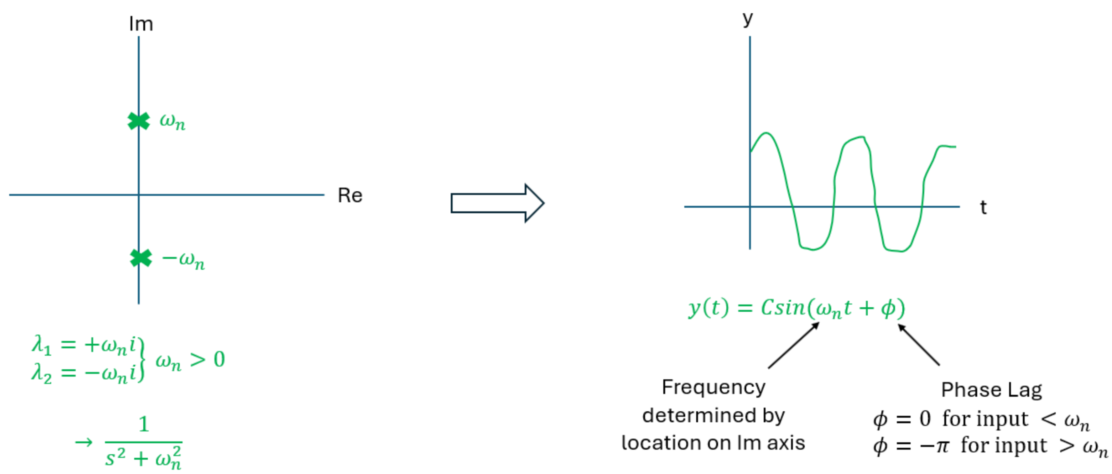

Figure 5.4: Second order system response for purely imaginary poles.

Case 3: λ_1, λ_2 complex with negative real part

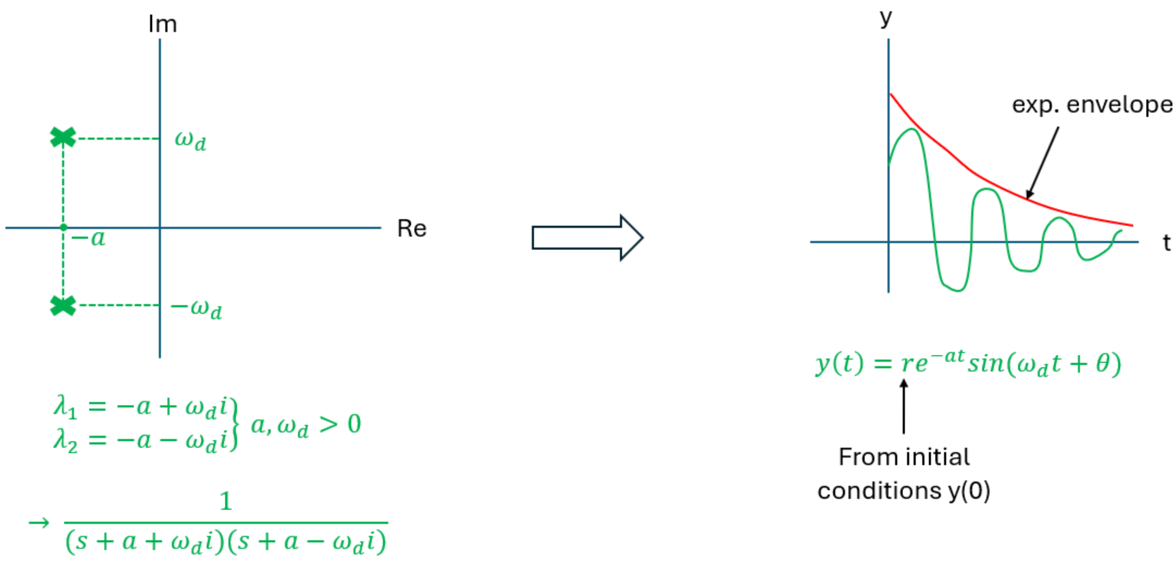

Figure 5.5: Second order system response for complex poles with negative real part.

Case 4: λ_1, λ_2 complex with positive real parts

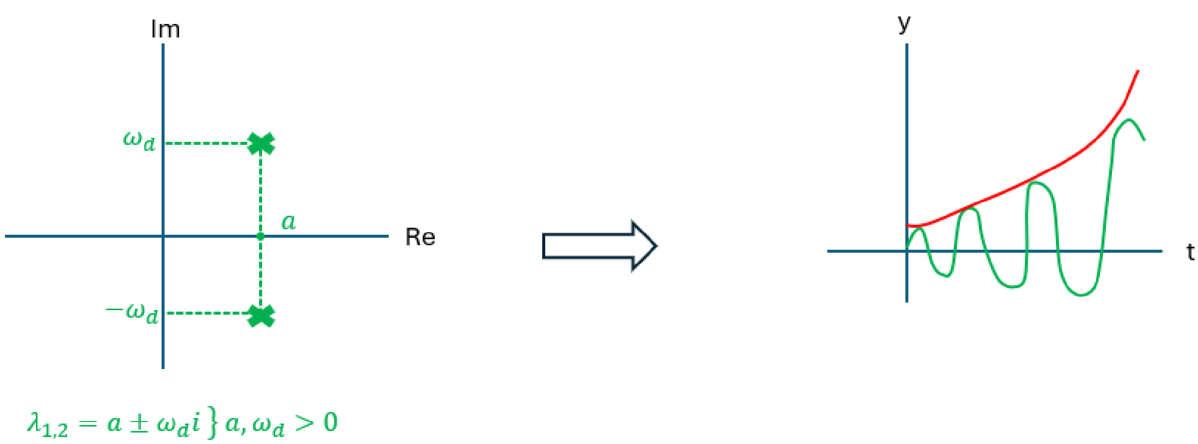

Figure 5.6: Second order system response for complex poles with positive real part.

Case 5: λ_1, λ_2 are real and at least one is positive

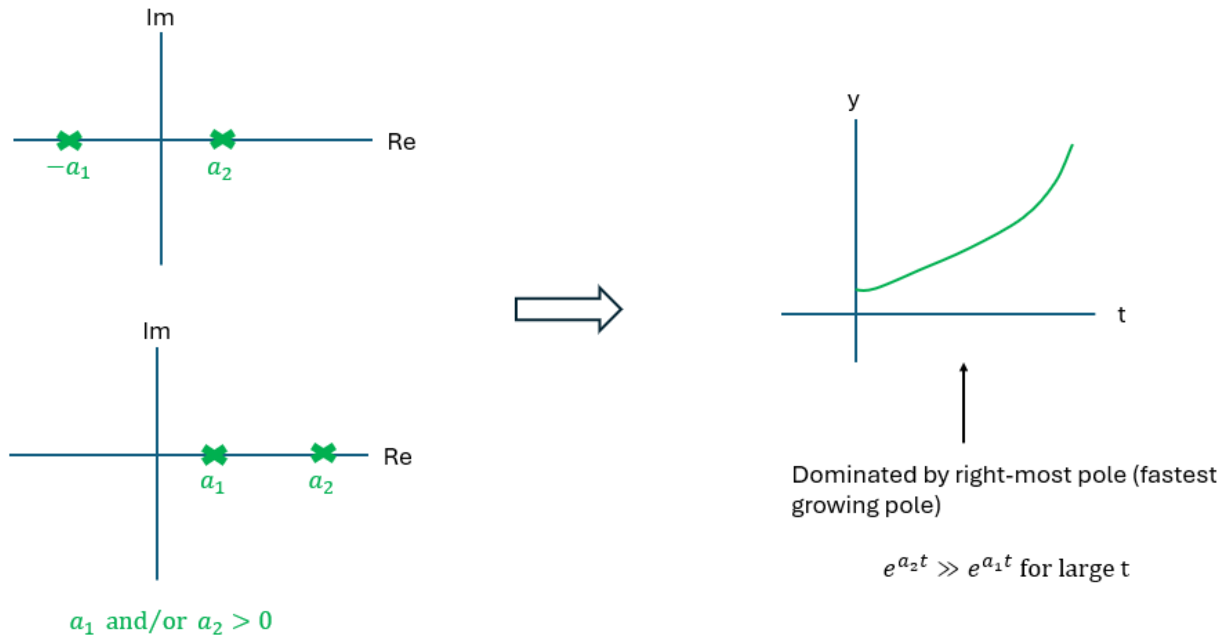

Figure 5.7: Second order system response for real poles with at least one being positive.

$\therefore$ For 2nd order systems, you can quickly check stability by looking at the coefficients of the characteristic equation, even without factoring:

$$\boxed{s^2 + a_1 s + a_2 = 0 \text{ has 2 solutions with } \underline{\text{negative real part (stable)}} \text{ iff } a_1, a_2 > 0}$$

5.3 Analysis of Stable 2nd Order Systems

Analysis is easier by defining variables $\zeta, \omega_n, \omega_d$, where ζ is the damping ratio, ω_n is the natural frequency, and ω_d is the damped natural frequency.

$$\rightarrow \quad \text{In time domain} : \ddot{y}(t) + \overbrace{2\zeta\omega_n}^{a_1} \dot{y}(t) + \overbrace{\omega_n^2}^{a_2} y(t) = b\omega_n^2 u(t)$$

$$\rightarrow \quad \text{In s-domain} : G(s) = \frac{Y}{U} = \frac{b\omega_n^2}{s^2 + \underbrace{2\zeta\omega_n}_{a_1} s + \underbrace{\omega_n^2}_{a_2}}$$

The roots of the characteristic equation are:

$$\lambda_1, \lambda_2 = -a \pm i\omega_d, \qquad \left.\begin{matrix} a = \zeta\omega_n \\ \omega_d = \omega_n\sqrt{1-\zeta^2} \end{matrix}\right\} \text{Valid for } \zeta \geq 0 \text{ (stable systems)}$$

Let's visualize this

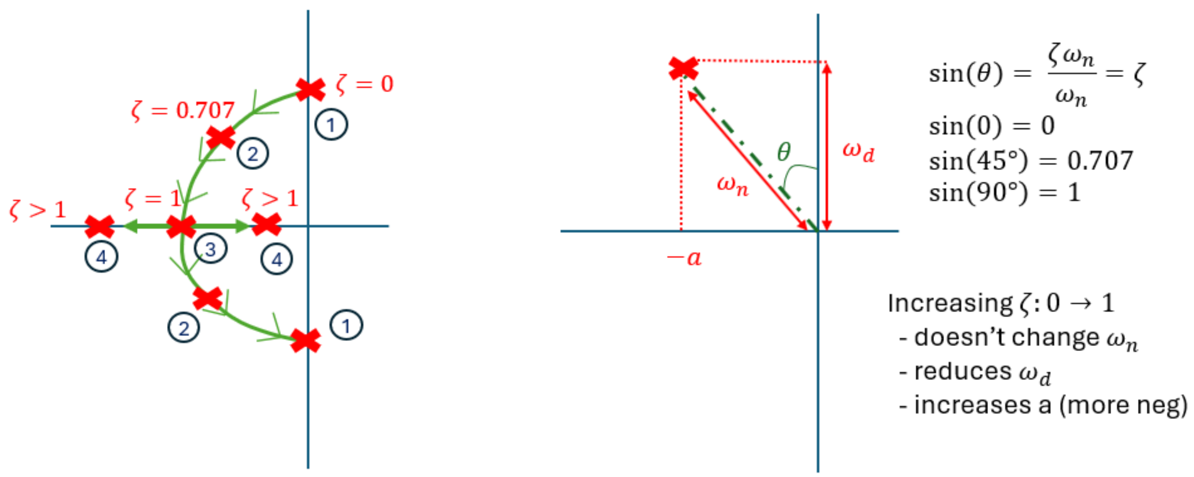

Figure 5.8: Damping ratio effect on poles for second-order system.

1. No damping

$$\text{If } \zeta = 0, \text{ then } \left.\begin{aligned} a &= 0 \\ \omega_d &= \omega_n \end{aligned}\right\} \rightarrow \lambda_1, \lambda_2 = 0 \pm i\omega_d$$

No exponential growth/decay since there is no damping.

Purely imaginary, just oscillates.

> Our characteristic equation becomes $(s^2 + 0 + \omega_n^2)$, which you might recognize as the Laplace Transform of $sin(\omega_n t)$

2. Underdamped

$$\text{If } 0 < \zeta < 1, \text{ then } \left.\begin{aligned} a &= \zeta\omega_n \\ \omega_d &\text{ is real} \end{aligned}\right\} \rightarrow \lambda_1, \lambda_2 = \overbrace{-a}^{R_e} \pm \overbrace{i\omega_d}^{Im}$$

Complex conjugates → decay + oscillation.

3. Critically Damped

$$\text{If } \zeta = 1, \text{ then } \left.\begin{aligned} a &= \omega_n \\ \omega_d &= 0 \end{aligned}\right\} \rightarrow \lambda_1, \lambda_2 = -a \pm 0$$

Purely real → just decay.

Repeated poles at -a

4. Overdamped

$$\text{If } \zeta > 1, \text{ then } \left.\begin{array}{l} a = \zeta\omega_n \\ \omega_d \text{ is imaginary} \end{array}\right\} \rightarrow \lambda_1, \lambda_2 = -a \pm \omega_n\sqrt{\zeta^2 - 1}$$

Real, distinct poles.

Notice $i\omega_d$ becomes real:

$$\begin{aligned} i\omega_d &= i\omega_n\sqrt{1-\zeta^2} \\ &= i\omega_n\sqrt{i^2(\zeta^2-1)} \\ &= i^2\omega_n\sqrt{\zeta^2-1} \\ &= -\omega_n\sqrt{\zeta^2-1} \end{aligned}$$

Question: Why might overdamping compromise performance, e.g. rise time?

Answer → Moves right most pole (which is dominant) further right (slow decay)

Example: Overdamped (Case 4)

Let's find the roots for a 2nd order polynomial using variables $\zeta, \omega_n, \omega_d$. The purpose of this exercise is to find the roots, but this time let's generate our polynomial from roots that we *do* know so we can trace and confirm the solution:

$$\begin{aligned} &(s+8)(s+2) = s^2 + \underbrace{10}_{2\zeta\omega_n} s + \underbrace{16}_{\omega_n^2} \\ &\omega_n = \sqrt{16} = 4 \\ &2\zeta\omega_n = 10 \quad \rightarrow \quad \zeta = \frac{10}{2 \cdot 4} = \frac{5}{4} \end{aligned}$$

Graphically:

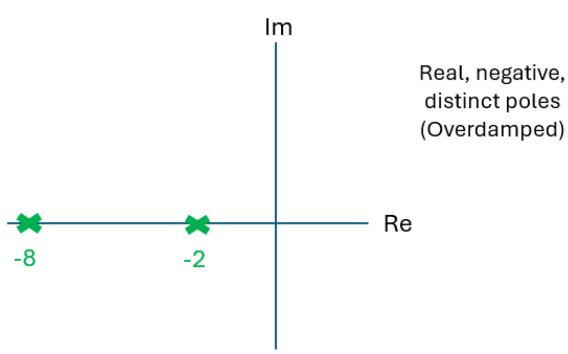

Figure 5.9: Overdamped example.

The roots of the characteristic equation are:

$$\lambda_1, \lambda_2 = -a \pm i\omega_d$$
$$\text{Solving for a and } \omega_d :$$
$$a = \zeta\omega_n = 5,$$
$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 4\sqrt{1-\frac{25}{16}}$$
$$= 4\sqrt{-\frac{9}{16}} = 4\sqrt{i^2\frac{9}{16}} = 4 \cdot i \cdot \frac{3}{4} = 3i$$

$$\implies \lambda_1, \lambda_2 = -5 \pm i \cdot 3i$$
$$= -5 \pm 3$$

$$\therefore \quad \lambda_1 = -8, \quad \lambda_2 = -2$$